Queueing analysis of a battery-backed electric vehicle fast charging system

Andres Ferragut

with Sebastián Bibiloni (Universidad de la República)

IFIP Performance 2026

Outline

Introduction

Single-server model

Single-server in steady-state

Optimal battery regime

Numerical results

The many-server case

Conclusions

Outline

Introduction

Single-server model

Single-server in steady-state

Optimal battery regime

Numerical results

The many-server case

Conclusions

Motivation

- DC fast chargers deliver 150 kW or more per port, but stations are idle a large portion of the time.
- Sizing the grid connection for the peak is wasteful: infrastructure costs, and demand charges may dominate operating costs [National Renewable Energy Laboratory, 2023].

Motivation

- DC fast chargers deliver 150 kW or more per port, but stations are idle a large portion of the time.
- Sizing the grid connection for the peak is wasteful: infrastructure costs, and demand charges may dominate operating costs [National Renewable Energy Laboratory, 2023].

Idea

Place a battery between the grid and the charger: charge it slowly from the grid, discharge it fast into the car.

Motivation

- DC fast chargers deliver 150 kW or more per port, but stations are idle a large portion of the time.
- Sizing the grid connection for the peak is wasteful: infrastructure costs, and demand charges may dominate operating costs [National Renewable Energy Laboratory, 2023].

Idea

Place a battery between the grid and the charger: charge it slowly from the grid, discharge it fast into the car.

- Battery storage at EV stations: mostly studied for renewable integration [Bayram et al., 2011; Kabir et al., 2020; Tran et al., 2019; Tang et al., 2026].
- Queueing models of EV charging: mostly fixed service rates [Bayram et al., 2013; Dupont and Hayel, 2023; Tan et al., 2014].

The question

A queueing question

How often does a vehicle find the battery depleted, and is forced to charge at the slower grid rate?

- The coupling between the continuous battery state and the discrete occupancy distinguishes this system from a standard queue.
- Natural framework: piecewise-deterministic Markov processes (PDMP) [Davis, 1984, 1993].

Key points of this talk

- A PDMP model of a battery-buffered station, with state (occupancy, battery level).
- Closed-form stationary distribution for the single-server loss system, with a decoupling property.
- Explicit blocking probability, mean charging time and slowdown factor.
- An optimal battery regime: average slowdown depends only on hardware parameters $\Rightarrow$ explicit battery sizing.
- The many-server case: an eigenvalue problem for a matrix pencil.

Outline

Introduction

Single-server model

Single-server in steady-state

Optimal battery regime

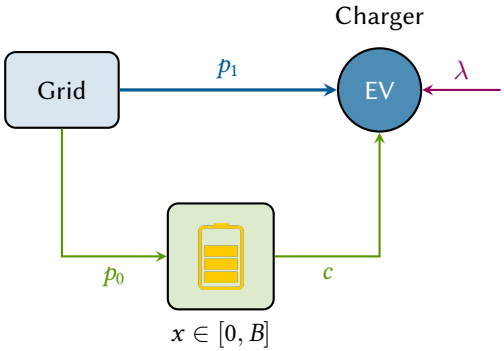
Numerical results

The many-server case

Conclusions

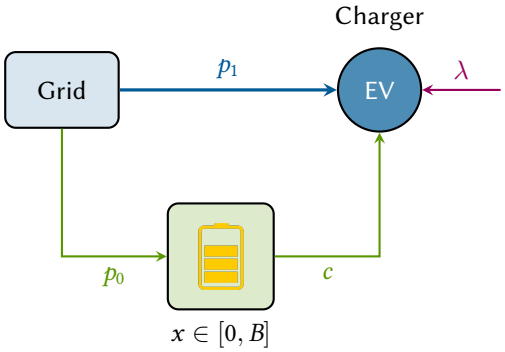
Setting

- Poisson arrivals λ, demands $Y \sim \text{Exp}(\mu)$.
- Loss system: one charger, cars finding it busy are lost.
- Battery level $x(t) \in [0, B]$.
- p_1: grid power; $p_2 > p_1$: fast charging; $c := p_2 - p_1$; $0 < p_0 \leq p_1$: idle recharge.



Setting

- Poisson arrivals λ, demands $Y \sim \text{Exp}(\mu)$.
- Loss system: one charger, cars finding it busy are lost.
- Battery level $x(t) \in [0, B]$.
- p_1: grid power; $p_2 > p_1$: fast charging; $c := p_2 - p_1$; $0 < p_0 \leq p_1$: idle recharge.
- Two-phase service: fast at p_2 while $x > 0$ (battery drains at c), slow at p_1 once $x = 0$.

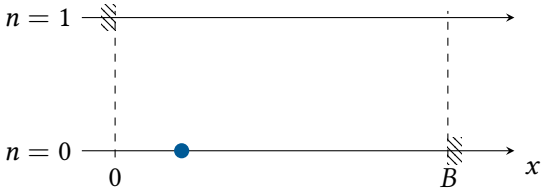


Departure rate $\mu p(x)$, with

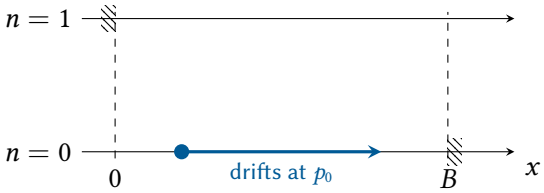
$$p(x) = p_2 \, \mathbf{1}_{\{x>0\}} + p_1 \, \mathbf{1}_{\{x=0\}}.$$

State space and dynamics

- State: $Z(t) = (n(t), x(t)) \in E = \{0, 1\} \times [0, B]$.



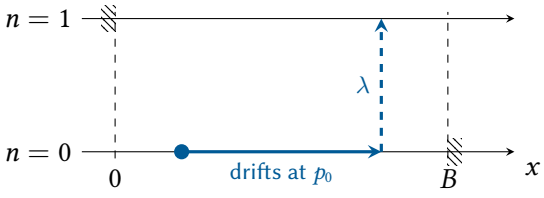
State space and dynamics



- State: $Z(t) = (n(t), x(t)) \in E = \{0, 1\} \times [0, B]$.
- Between jumps $\dot{x} = r(n, x)$:

$$r(n,x) = \begin{cases} p_0 & n = 0,\ x < B, \\ 0 & n = 0,\ x = B, \\ -c & n = 1,\ x > 0, \\ 0 & n = 1,\ x = 0. \end{cases}$$

State space and dynamics

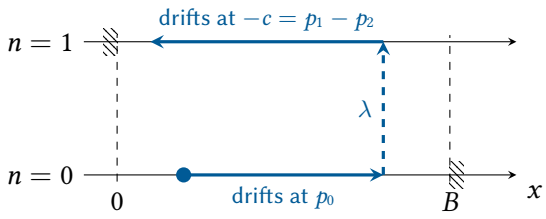


- State: $Z(t) = (n(t), x(t)) \in E = \{0, 1\} \times [0, B]$.
- Between jumps $\dot{x} = r(n, x)$:

$$r(n,x) = \begin{cases} p_0 & n = 0,\ x < B, \\ 0 & n = 0,\ x = B, \\ -c & n = 1,\ x > 0, \\ 0 & n = 1,\ x = 0. \end{cases}$$

- Jumps: $(0, x) \mapsto (1, x)$ at rate λ,

State space and dynamics

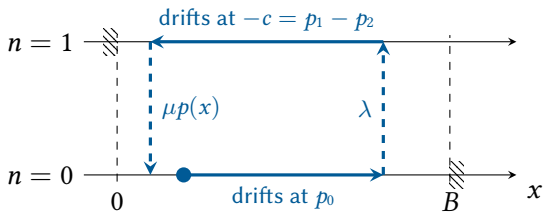


- State: $Z(t) = (n(t), x(t)) \in E = \{0, 1\} \times [0, B]$.
- Between jumps $\dot{x} = r(n, x)$:

$$r(n,x) = \begin{cases} p_0 & n = 0,\ x < B, \\ 0 & n = 0,\ x = B, \\ -c & n = 1,\ x > 0, \\ 0 & n = 1,\ x = 0. \end{cases}$$

- Jumps: $(0, x) \mapsto (1, x)$ at rate λ,

State space and dynamics



- State: $Z(t) = (n(t), x(t)) \in E = \{0, 1\} \times [0, B]$.
- Between jumps $\dot{x} = r(n, x)$:

$$r(n,x) = \begin{cases} p_0 & n = 0,\ x < B, \\ 0 & n = 0,\ x = B, \\ -c & n = 1,\ x > 0, \\ 0 & n = 1,\ x = 0. \end{cases}$$

- Jumps: $(0, x) \mapsto (1, x)$ at rate λ, $(1, x) \mapsto (0, x)$ at rate $\mu p(x)$.
- The battery level is unchanged at jumps.

A PDMP outside the standard framework

The standard theory [Davis, 1984, 1993] requires an open state space and forces a jump before reaching the boundary. Here E is compact, and the flow reaches $x = 0$, $x = B$ in finite time.

- The vector field is bounded and monotone on each mode $\Rightarrow$ well-defined flow.
- Bounded jump rates, jumps leave x unchanged $\Rightarrow$ strong Markov via the martingale problem (cf. Davis [1984, Thm. 4.8]).

A PDMP outside the standard framework

The standard theory [Davis, 1984, 1993] requires an open state space and forces a jump before reaching the boundary. Here E is compact, and the flow reaches $x = 0$, $x = B$ in finite time.

- The vector field is bounded and monotone on each mode $\Rightarrow$ well-defined flow.
- Bounded jump rates, jumps leave x unchanged $\Rightarrow$ strong Markov via the martingale problem (cf. Davis [1984, Thm. 4.8]).

Extended generator, for $\psi(n, \cdot)$ absolutely continuous on $[0, B]$:

$$\mathcal{A}\psi(0, x) = r(0, x)\, \psi_0'(x) + \lambda\, [\psi(1, x) - \psi(0, x)],$$
$$\mathcal{A}\psi(1, x) = r(1, x)\, \psi_1'(x) + \mu p(x)\, [\psi(0, x) - \psi(1, x)].$$

A stationary distribution satisfies $\int \mathcal{A}\psi \, d\pi = 0$ for all $\psi \in \mathcal{D}(\mathcal{A})$: the weak form of $\mathcal{A}^* \pi = 0$.

Outline

Introduction

Single-server model

Single-server in steady-state

Optimal battery regime

Numerical results

The many-server case

Conclusions

Balance equations

E compact, $Z(t)$ irreducible $\Rightarrow$ unique π [Meyn and Tweedie, 1993]. We seek a mixed measure:

$$\pi = f_0(x)\,dx + f_1(x)\,dx + \pi_{0,B}\,\delta_{(0,B)} + \pi_{1,0}\,\delta_{(1,0)},$$

with atoms at the corners where the drift points towards the boundary.

Balance equations

E compact, $Z(t)$ irreducible $\Rightarrow$ unique π [Meyn and Tweedie, 1993]. We seek a mixed measure:

$$\pi = f_0(x)\,dx + f_1(x)\,dx + \pi_{0,B}\,\delta_{(0,B)} + \pi_{1,0}\,\delta_{(1,0)},$$

with atoms at the corners where the drift points towards the boundary.

Interior $x \in (0, B)$:

$$\begin{aligned} p_0 f_0'(x) &= -\lambda f_0(x) + \mu p_2 f_1(x), \\ -c f_1'(x) &= \lambda f_0(x) - \mu p_2 f_1(x). \end{aligned}$$

Boundary fluxes:

$$\begin{aligned} c f_1(0^+) &= \mu p_1 \pi_{1,0} = p_0 f_0(0^+), \\ p_0 f_0(B^-) &= \lambda \pi_{0,B} = c f_1(B^-). \end{aligned}$$

Stationary distribution

Theorem

Let $\theta := \dfrac{\mu p_2}{c} - \dfrac{\lambda}{p_0}$. *The unique stationary distribution of* $Z(t)$ *is:*

$$f_0(x) = \frac{e^{\theta x}}{K}, \qquad f_1(x) = \frac{p_0}{c} \cdot \frac{e^{\theta x}}{K}, \qquad x \in (0, B),$$

$$\pi_{1,0} = \frac{p_0}{\mu p_1 K}, \qquad \pi_{0,B} = \frac{p_0\, e^{\theta B}}{\lambda K},$$

with

$$K = \frac{p_0 + c}{c} \cdot \frac{e^{\theta B} - 1}{\theta} + \frac{p_0}{\mu p_1} + \frac{p_0\, e^{\theta B}}{\lambda}.$$

Proof sketch: decoupling

- Adding both interior equations: $\frac{d}{dx}\left[p_0 f_0 - c f_1\right] = 0$.
- The flux conditions at $x = 0$ give $p_0 f_0(0^+) = c f_1(0^+)$: the constant is zero.

Decoupling property

$$f_1(x) = \frac{p_0}{c} f_0(x) \quad \forall x \in (0, B) :$$

charger occupancy and battery level are independent given the battery is in the interior.

Proof sketch: decoupling

- Adding both interior equations: $\dfrac{d}{dx}\left[p_0 f_0 - c f_1\right] = 0$.
- The flux conditions at $x = 0$ give $p_0 f_0(0^+) = c f_1(0^+)$: the constant is zero.

Decoupling property

$$f_1(x) = \frac{p_0}{c} f_0(x) \quad \forall x \in (0, B) :$$

charger occupancy and battery level are independent given the battery is in the interior.

- Substituting: $f_0' = \theta f_0$, so the densities are exponential.
- Atoms follow from the boundary equations.

Interpretation of θ

In an idle-busy cycle ignoring the boundaries, the expected net battery change is

$$\Delta := \frac{p_0}{\lambda} - \frac{c}{\mu p_2} \qquad \text{(same sign as } \theta\text{)}.$$

Interpretation of θ

In an idle-busy cycle ignoring the boundaries, the expected net battery change is

$$\Delta := \frac{p_0}{\lambda} - \frac{c}{\mu p_2} \qquad \text{(same sign as } \theta\text{)}.$$

With offered load $\rho := \lambda/\mu$ (average external power request):

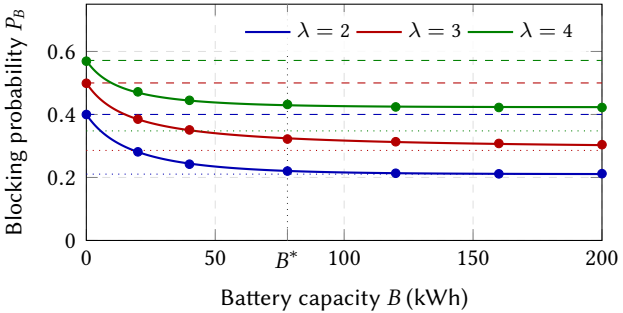
- $\theta > 0 \Leftrightarrow \rho < \dfrac{p_0 p_2}{c}$: lightly loaded, the battery tends to fill.
- $\theta < 0$: highly loaded, the battery tends to drain, much service at the slow rate p_1.
- $\theta = 0$: the battery neither fills nor drains on average, uniform density on $(0, B)$.

Blocking probability

By PASTA, P_B is the probability that the charger is working:

$$P_B = \int_0^B f_1(x)\,dx + \pi_{1,0} = \frac{p_0}{K}\left(\frac{e^{\theta B}-1}{c\,\theta} + \frac{1}{\mu p_1}\right).$$

- $B \to 0$: M/M/1/1 at rate p_1, $P_B \to \frac{\rho}{\rho+p_1}$.
- $B \to \infty$, $\theta > 0$: M/M/1/1 at rate p_2, $P_B \to \frac{\rho}{\rho+p_2}$.
- $B \to \infty$, $\theta < 0$: strictly in between, the battery regularly depletes.



Dashed/dotted: M/M/1/1 at p_1/p_2. Dots: simulation.

Charging time and average slowdown

At most one car in service, so $E[N] = P_B$. By Little's law:

$$E[T] = \frac{P_B}{\lambda(1-P_B)} = \frac{p_0\left(\dfrac{e^{\theta B}-1}{c\,\theta} + \dfrac{1}{\mu p_1}\right)}{\lambda\,\dfrac{e^{\theta B}-1}{\theta} + p_0\,e^{\theta B}}.$$

- Average slowdown w.r.t. always serving at p_2: $\bar{S} := \dfrac{\mu p_2\,E[T]}{E[Y]}$.
- In the balanced case $\theta = 0$: $E[T] = \dfrac{1}{1+\lambda B/p_0}\left(\dfrac{B}{c} + \dfrac{1}{\mu p_1}\right)$.

Slowdown factor

Service degradation as a function of the requested energy:

$$S(y) := \frac{E[T \mid Y = y]}{y/p_2} \geq 1.$$

- Key quantity: $x_y := cy/p_2$, the battery level exactly depleted by demand y.
- A car with demand y admitted at level x takes

$$T(x, y) = \frac{y}{p_2} + \frac{(x_y - x)^+}{p_1}.$$

- For $y \geq x^* = p_2 B/c$, slow charging is unavoidable regardless of the battery level.

Slowdown factor

Closed form

Proposition

Let $\mathcal{J}(a) = \dfrac{e^{\theta a} - 1 - \theta a}{\theta^2}$ *and* $K(1 - P_B) = \dfrac{e^{\theta B} - 1}{\theta} + \dfrac{p_0 e^{\theta B}}{\lambda}$. *Then:*

$$S(y) = 1 + \frac{p_2}{p_1\, y} \cdot \frac{\mathcal{J}(x_y)}{K(1 - P_B)}, \qquad y < \frac{p_2 B}{c},$$

$$S(y) = 1 + \frac{p_2}{p_1\, y} \left[\frac{\mathcal{J}(B)}{K(1 - P_B)} + (x_y - B) \right], \qquad y \geq \frac{p_2 B}{c}.$$

Slowdown factor

Closed form

Proposition

Let $\mathcal{J}(a) = \dfrac{e^{\theta a} - 1 - \theta a}{\theta^2}$ *and* $K(1 - P_B) = \dfrac{e^{\theta B} - 1}{\theta} + \dfrac{p_0 e^{\theta B}}{\lambda}$. *Then:*

$$S(y) = 1 + \frac{p_2}{p_1\, y} \cdot \frac{\mathcal{J}(x_y)}{K(1 - P_B)}, \qquad y < \frac{p_2 B}{c},$$

$$S(y) = 1 + \frac{p_2}{p_1\, y} \left[\frac{\mathcal{J}(B)}{K(1 - P_B)} + (x_y - B) \right], \qquad y \geq \frac{p_2 B}{c}.$$

- $S(y) \to 1$ as $y \to 0$; $S(y) \to p_2/p_1$ as $y \to \infty$ or $B \to 0$.
- As $B \to \infty$: $S(y) \to 1$ only if $\theta \geq 0$. If $\theta < 0$, $S(y) \to 1 + \frac{p_2}{p_1 y} |\theta| \mathcal{J}(x_y) > 1$: even an infinite battery does not remove the slowdown.

Outline

Introduction

Single-server model

Single-server in steady-state

Optimal battery regime

Numerical results

The many-server case

Conclusions

Design criterion

For battery health: charge it slowly, and keep it cycling rather than fully charged or depleted for long periods.

Design criterion

Choose the idle recharge rate p_0 to reach the balanced regime $\theta = 0$.

Design criterion

For battery health: charge it slowly, and keep it cycling rather than fully charged or depleted for long periods.

Design criterion

Choose the idle recharge rate p_0 to reach the balanced regime $\theta = 0$.

- Setting $\theta = 0$ uniquely determines

$$p_0^* = \frac{\lambda c}{\mu p_2} = \tilde{\rho}\, c, \qquad \tilde{\rho} := \frac{\rho}{p_2} = \frac{\lambda}{\mu p_2}.$$

- Feasible ($p_0^* \le p_1$) iff $\rho \le \rho_{\max} = p_1 p_2 / c$. Above it, $\theta < 0$ regardless of p_0.

Average slowdown in the optimal regime

- $\alpha = p_2/p_1 > 1$: speed-up ratio.
- $\beta = \mu B p_2/c$: normalized battery capacity, number of fast-charge sessions the battery sustains.

Average slowdown in the optimal regime

- $\alpha = p_2/p_1 > 1$: speed-up ratio.
- $\beta = \mu B p_2 / c$: normalized battery capacity, number of fast-charge sessions the battery sustains.

With $\theta = 0$ and $p_0 = p_0^*$:

$$\bar{S} = \frac{\alpha + \beta}{\beta + 1}.$$

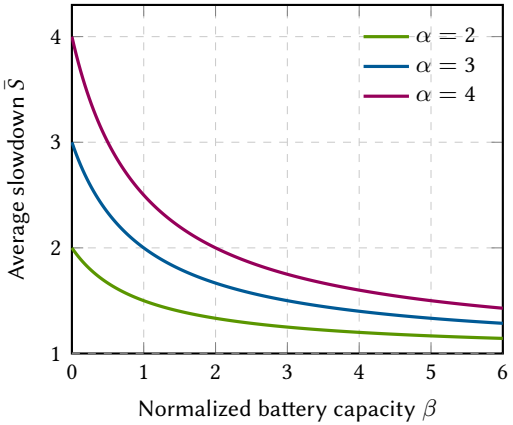
Average slowdown in the optimal regime

- $\alpha = p_2/p_1 > 1$: speed-up ratio.
- $\beta = \mu B p_2 / c$: normalized battery capacity, number of fast-charge sessions the battery sustains.

With $\theta = 0$ and $p_0 = p_0^*$:

$$\bar{S} = \frac{\alpha + \beta}{\beta + 1}.$$

Remarkably, it does not depend on the load: the same design works for all $\rho < \rho_{\max}$!



Battery sizing

Setting a target slowdown $\bar{S}_0 \in (1, \alpha)$:

$$\beta^* = \frac{\alpha - \bar{S}_0}{\bar{S}_0 - 1} \quad \Longleftrightarrow \quad B^* = \frac{c}{\mu p_2} \cdot \frac{\alpha - \bar{S}_0}{\bar{S}_0 - 1}.$$

- β^* grows with α: a large speed-up drains the battery fast relative to its recharge.

Battery sizing

Setting a target slowdown $\bar{S}_0 \in (1, \alpha)$:

$$\beta^* = \frac{\alpha - \bar{S}_0}{\bar{S}_0 - 1} \quad \Longleftrightarrow \quad B^* = \frac{c}{\mu p_2} \cdot \frac{\alpha - \bar{S}_0}{\bar{S}_0 - 1}.$$

- β^* grows with α: a large speed-up drains the battery fast relative to its recharge.

Blocking probability in this regime:

$$P_B = \frac{\tilde{\rho}\bar{S}_0}{1 + \tilde{\rho}\bar{S}_0},$$

an M/M/1/1 blocking formula with effective load $\tilde{\rho}\bar{S}_0$.

Outline

Introduction

Single-server model

Single-server in steady-state

Optimal battery regime

Numerical results

The many-server case

Conclusions

A design example

Hardware:

- $p_2 = 150$ kW: typical DC fast charger.
- $p_1 = 60$ kW: medium-sized grid connection.
- $c = 90$ kW, $\alpha = 2.5$.

Demand:

- $1/\mu = 20$ kWh (22 kWh average per DC fast session [National Renewable Energy Laboratory, 2023]).
- Mean fast session: $1/(\mu p_2) \approx 8$ min.
- Target $\lambda = 3$ cars/h.

A design example

Hardware:

- $p_2 = 150$ kW: typical DC fast charger.
- $p_1 = 60$ kW: medium-sized grid connection.
- $c = 90$ kW, $\alpha = 2.5$.

Demand:

- $1/\mu = 20$ kWh (22 kWh average per DC fast session [National Renewable Energy Laboratory, 2023]).
- Mean fast session: $1/(\mu p_2) \approx 8$ min.
- Target $\lambda = 3$ cars/h.

Design:

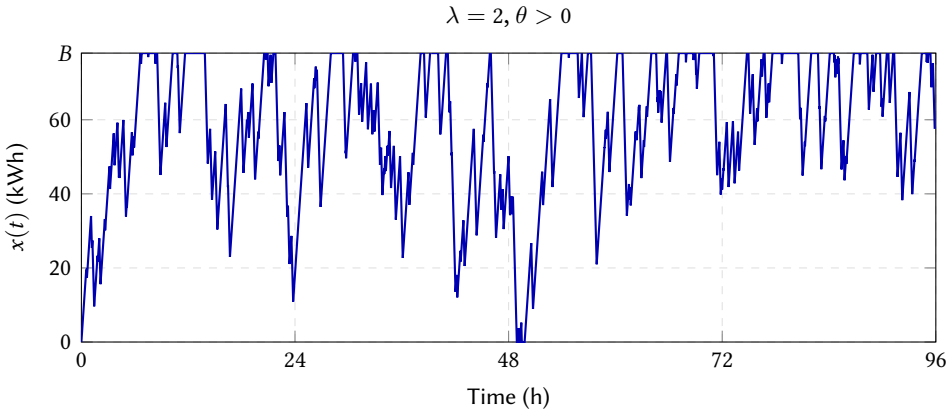
- $\theta = 0$: $p_0^* = \tilde{\rho} c = 0.4 \times 90 = 36$ kW.
- Feasible: $\tilde{\rho} = 0.4 < 1/(\alpha - 1) = 2/3$.
- Target $\bar{S}_0 = 1.2$: $\beta^* = 6.5$, so

$$B^* = 78 \text{ kWh}.$$

Mean service time: $8 \times 1.2 = 9.6$ min, vs. 20 min at the slow rate p_1.

Sample paths

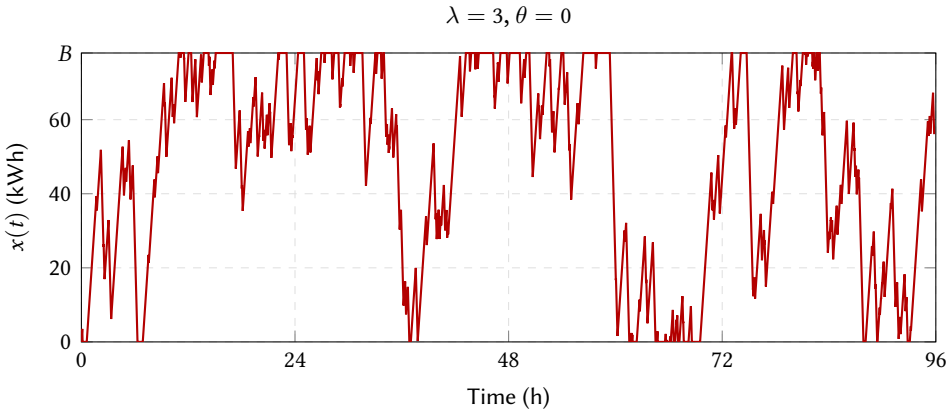
Three load regimes, same design



Lightly loaded: the battery drifts towards B and spends long periods near full.

Sample paths

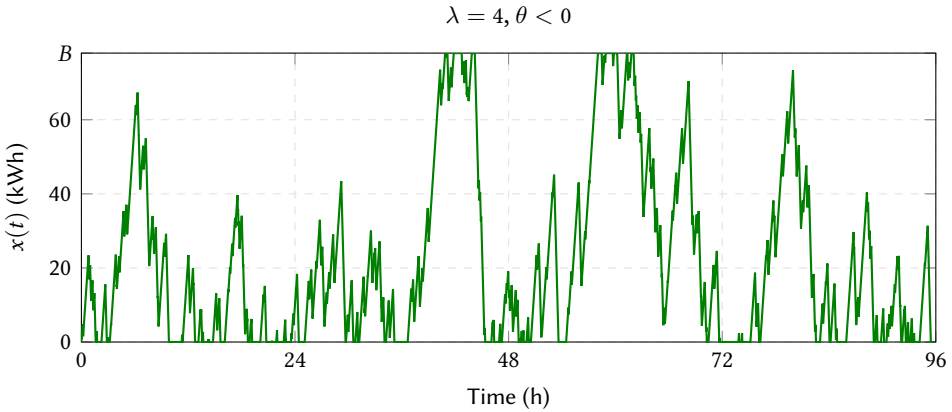
Three load regimes, same design



Critical: the battery oscillates freely across $(0, B)$ without dwelling at either boundary.

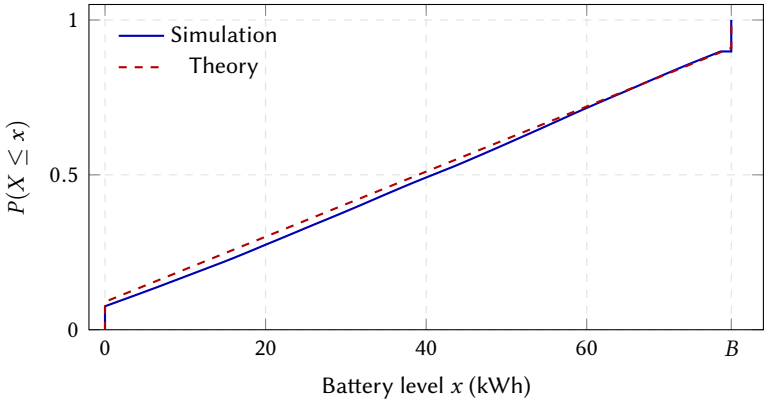
Sample paths

Three load regimes, same design



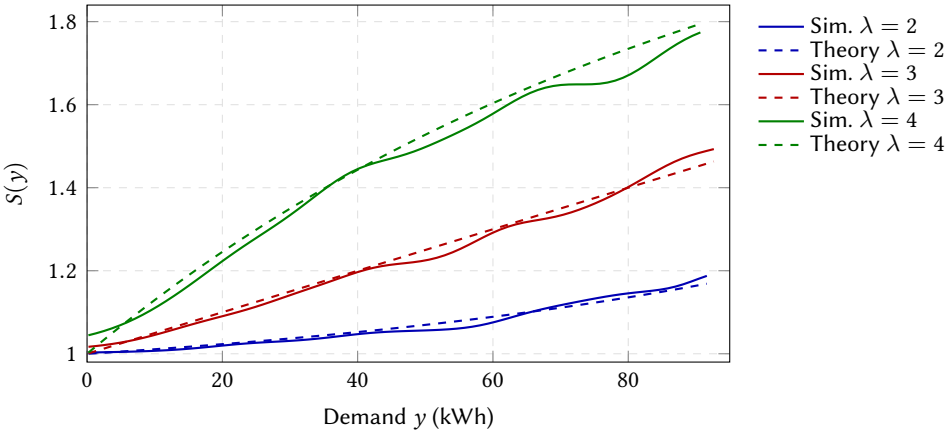
Heavily loaded: the battery drains, and service frequently degrades to the slow rate.

Stationary distribution of the battery level



$\lambda = 3$ ($\theta = 0$): uniform density on $(0, B)$ plus atoms at $x = 0$ and $x = B$ ($\pi_{1,0} = \pi_{0,B} = 0.0901$).

Slowdown factor



Solid: kernel regression from simulation. Dashed: analytical formula.

Outline

Introduction

Single-server model

Single-server in steady-state

Optimal battery regime

Numerical results

The many-server case

Conclusions

Several chargers: a matrix pencil

s chargers share the battery and the grid ($p_0 = p_1$). Drift $r_n = p_1 - np_2$: charges for $n \le n^* := \lfloor p_1/p_2 \rfloor$, drains above.

- Interior balance: $R\,\mathbf{f}'(x) = Q^T\,\mathbf{f}(x)$, $R = \mathrm{diag}(r_0, \dots, r_s)$, Q the occupancy chain generator.
- R invertible when $p_1/p_2 \notin \mathbb{N}$: solutions $\mathbf{a}\, e^{\theta x}$ with $(\theta, \mathbf{a})$ an eigenpair of $R^{-1}Q^T$.
- $\theta = 0$ is always an eigenvalue, with Erlang-B weights $a_n = \tilde{\rho}^{\,n}/n!$.

Several chargers: a matrix pencil

s chargers share the battery and the grid ($p_0 = p_1$). Drift $r_n = p_1 - np_2$: charges for $n \le n^* := \lfloor p_1/p_2 \rfloor$, drains above.

- Interior balance: $R\,\mathbf{f}'(x) = Q^T\,\mathbf{f}(x)$, $R = \mathrm{diag}(r_0, \dots, r_s)$, Q the occupancy chain generator.
- R invertible when $p_1/p_2 \notin \mathbb{N}$: solutions $\mathbf{a}\, e^{\theta x}$ with $(\theta, \mathbf{a})$ an eigenpair of $R^{-1}Q^T$.
- $\theta = 0$ is always an eigenvalue, with Erlang-B weights $a_n = \tilde{\rho}^{\,n}/n!$.

- The interior solution superposes all $s+1$ modes; the boundary conditions give exactly $s+1$ equations.
- Unlike $s = 1$, the full spectrum of the pencil (Q, R) is generically needed.

Ongoing work: the eigenvalues are always real (reversibility of Q), and a solution with positive densities and corner atoms exists and is unique.

Outline

Introduction

Single-server model

Single-server in steady-state

Optimal battery regime

Numerical results

The many-server case

Conclusions

Messages from the talk

- A battery-buffered EV fast charging station is naturally a PDMP.
- Single-server: closed-form stationary distribution, with a decoupling property.
- Explicit blocking probability, mean charging time and slowdown factor.
- Optimal regime ($\theta = 0$): average slowdown depends only on hardware, not on load $\Rightarrow$ explicit battery sizing.
- Many servers: all the eigenmodes of the pencil (Q, R) are needed.

Thank you!

Andres Ferragut
ferragut@ort.edu.uy
https://aferragu.github.io

References I

I. S. Bayram, G. Michailidis, M. Devetsikiotis, S. Bhattacharya, A. Chakrabortty, and F. Granelli. Local energy storage sizing in plug-in hybrid electric vehicle charging stations under blocking probability constraints. In *Proc. IEEE International Conference on Smart Grid Communications (SmartGridComm)*, pages 78–83, 2011.

I. S. Bayram, G. Michailidis, M. Devetsikiotis, and F. Granelli. Electric power allocation in a network of fast charging stations. *IEEE Journal on Selected Areas in Communications*, 31(7): 1235–1246, 2013.

M. H. A. Davis. Piecewise-deterministic Markov processes: a general class of non-diffusion stochastic models. *Journal of the Royal Statistical Society: Series B*, 46(3):353–388, 1984.

M. H. A. Davis. *Markov Models and Optimization*. Chapman & Hall, London, 1993.

A. Dupont and Y. Hayel. A loss queueing game for electric vehicle charging performance evaluation. *Performance Evaluation*, 161:102365, 2023.

M. E. Kabir, C. Assi, M. H. K. Tushar, and J. Yan. Optimal scheduling of EV charging at a solar power-based charging station. *IEEE Systems Journal*, 14(3):4221–4231, 2020.

S. P. Meyn and R. L. Tweedie. *Markov Chains and Stochastic Stability*. Springer, London, 1993.

References II

National Renewable Energy Laboratory. Battery energy storage for electric vehicle charging stations. Technical report, U.S. Department of Energy, 2023. Available at driveelectric.gov.

X. Tan, B. Sun, and D. H. K. Tsang. Queueing network models for electric vehicle charging station with battery swapping. In *Proc. IEEE International Conference on Smart Grid Communications (SmartGridComm)*, pages 1–6, 2014. doi: 10.1109/SmartGridComm.2014.7007613.

M. Tang, M. Wang, H. Li, Z. Tao, and W. Sheng. Coordinated ev charging scheduling strategy incorporating wind power integration and user demand. *Electric Power Systems Research*, 259: 113220, 2026.

V. T. Tran, M. R. Islam, K. M. Muttaqi, and D. Sutanto. An efficient energy management approach for a solar-powered EV battery charging facility to support distribution grids. *IEEE Transactions on Industry Applications*, 55(6):6517–6526, 2019.